

Access Database Basics — Detailed Lecture Notes

JKSSB Computer Awareness

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28.1 What a database is

A **database** is an organised collection of related data, stored so that it can be searched, updated, and retrieved easily. The data is not kept as a loose pile of separate files; it is arranged in a planned structure. A simple example is a school's information: the list of students, their classes, and their marks all belong together as related information, and a database keeps them that way.

A useful comparison is a spreadsheet. A spreadsheet (such as Excel) is built for calculation on a simple grid. A database is built for storing and linking large amounts of related data, and it can hold many tables that are related to one another. When a record system grows large, a database keeps the data consistent and avoids unnecessary duplication.

Definition (Database). *An organised collection of related data, stored so that it can be searched, updated, and retrieved easily.*

28.2 Database vs DBMS

A database and a DBMS are not the same thing, and the difference is a common exam point. The **database** is the collection of data itself. A **DBMS** (Database Management System) is the software that manages that data. The database is data; the DBMS is a program.

Point	Database	DBMS
What it is	The organised collection of data	The software that manages the data
Nature	Data	Program
Example	A school's records	MS Access, Oracle, MySQL

Definition (DBMS (Database Management System)). *The software that stores a database and lets users insert, update, delete, and retrieve data while controlling access and keeping the data consistent.*

A **relational database management system** (RDBMS) is a DBMS that keeps data in related tables. Microsoft Access is an example of an RDBMS.

28.3 What a DBMS does

A DBMS performs several jobs. It *stores* the data and controls how it is kept on disk. It lets users *insert*, *update*, *delete*, and *retrieve* records. It controls *access*, so that only permitted users can read or change the data. It also keeps the data *consistent* — different parts of the database are not allowed to contradict each other — and protects it from corruption.

Common DBMS software includes MS Access, Oracle, MySQL, and SQL Server. For this course, the important point is that Access is a DBMS: it is the program that manages a database, not the data itself.

28.4 Microsoft Access in the Office suite

Microsoft Access is the database program of the Microsoft Office suite. In the suite, Word handles text documents, Excel handles spreadsheets, PowerPoint handles presentations, and Access handles databases. Access is a relational DBMS: it stores its data in tables, and it also holds the queries, forms, and reports that work on that data. All of these objects are kept inside a single database file for each database.

Example 28.4.1. *A student is asked, “Which MS Office program is used to store and manage databases?” The answer is **Access**. Excel is a spreadsheet program, Word is a word processor, and PowerPoint is for presentations; none of them is a DBMS.*

28.5 The objects in an Access database

An Access database contains several kinds of object, each with one job.

Object	Job
Table	Stores the actual data in rows and columns
Query	Asks a question and retrieves selected data
Form	Provides a screen to enter and view data
Report	Presents data in a printable layout

The data itself always lives in **tables**. The other objects work on that data: a query retrieves it, a form is used to enter and view it, and a report presents it for printing.

28.6 Tables, records, and fields

A **table** is the basic object where data is stored. Inside a table, data is arranged in rows and columns. Each **column** is one **field** — one kind of information, such as Name or Marks. Each **row** is one **record** — one complete entry, such as the full details of one student. A table therefore holds many records, and every record has the same set of fields.

Term	Meaning	Position in a table
Field	One kind of information	A column
Record	One complete entry	A row
Table	A collection of related records	The whole grid

Definition (Record). *One complete entry in a table, held in a single row and made up of one value for each field.*

Definition (Field). *One kind of information in a table, held in a single column; each field has a name and a data type.*

Example 28.6.1. *In a Students table the columns are RollNo, Name, Marks, and Passed. One row — for example, “1, Ayaan, 78, Yes” — is one record. The Name column is the Name field. The whole set of rows together is the table.*

28.7 Data types in Access

Every field has a **data type**, which is the kind of value it can hold. Common Access data types are:

Data type	Used for
Short Text	Names, codes, and other text
Number	Values used in calculations
Date/Time	Dates and times
Yes/No	Only True or False (a Boolean value)
AutoNumber	A unique number added automatically for each new record

Because a field accepts only values that match its data type, the data stays clean. A field set to Number will not accept ordinary text, and a Yes/No field accepts only a true-or-false value.

28.8 Field properties and validation

A **field property** is a setting that describes how a field behaves. Some important properties are **Field Size**, which limits how much a field can hold; **Default Value**, which fills in a value automatically; **Required**, which, when set to Yes, stops the field from being left blank; and **Input Mask**, which fixes the pattern of entry, such as the format of a phone number.

Validation means checking that entered data is sensible and allowed. A **Validation Rule** states the condition a value must meet. For example, the rule **Between 0 And 100** accepts only values in that range. A **Validation Text** is the message shown when the rule is violated.

Definition (Validation). *The checking of entered data against a rule so that only sensible, allowed values are stored in a field.*

Example 28.8.1. *On a Marks field, the Validation Rule **Between 0 And 100** is set. A user types 150. Access rejects the entry because it fails the rule, and shows the Validation Text as a message. A valid value such as 78 is accepted normally.*

28.9 Primary key and index

A **primary key** is a field that uniquely identifies each record. Its value is different for every record and is never left blank. In the Students table, RollNo can serve as the primary key because no two students share a roll number. An AutoNumber field is often used as a ready-made primary key, because Access fills it with a fresh unique number for every new record.

An **index** is a special structure that makes searching and sorting a table faster. It works like the index of a book: instead of reading every page, it points straight to the right place. A primary key is automatically indexed. An index takes extra space and can slightly slow down the work of adding data, so it is created only where it is genuinely useful.

Definition (Primary key). *A field, or set of fields, whose value uniquely identifies every record in a table; it cannot contain duplicates and cannot be left blank.*

28.10 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. A **database** is the data; a **DBMS** is the software that manages it. Do not swap them.
2. A **field** is a column; a **record** is a row; a **table** is the whole collection of related records.
3. **Validation** checks the value entered; an **index** speeds up searching. They do different jobs.
4. A **primary key** is unique and cannot be blank; it cannot contain duplicate values.
5. Microsoft **Access** is a DBMS and part of the MS Office suite; it is not a spreadsheet.

Example 28.10.1. *An item asks which object stores the actual data in an Access database. Trace the objects: a query retrieves data, a form is for entry, and a report is for printing, but only the **table** stores the data. If the stem instead asks which field uniquely identifies a record, the answer is the **primary key**; if it asks which field setting stops a blank entry, the answer is **Required**.*

Exercises

1. Define a database and explain how it differs from a DBMS.
2. Distinguish a field, a record, and a table, stating the position of each in a table.
3. Name any four Access data types and give one use of each.
4. Explain the difference between a validation rule and an index.
5. What is a primary key, and why can it not contain duplicate values?

Solutions

1. A database is an organised collection of related data. A DBMS is the software that manages that data — it stores it, allows insert, update, delete, and retrieve operations, and controls access. The database is the data; the DBMS is the program.
2. A field is one kind of information and is a column in a table. A record is one complete entry and is a row in a table. A table is the whole collection of related records, arranged in rows and columns.
3. Any four of: Short Text (names and codes), Number (values used in calculations), Date/Time (dates and times), Yes/No (True or False), AutoNumber (a unique number for each new record).
4. A validation rule checks that entered data meets a condition, such as **Between 0 And 100**, and rejects values that fail it. An index is a structure that speeds up searching and sorting; it does not check values.
5. A primary key is a field whose value uniquely identifies each record. It cannot contain duplicates because two records with the same key value could not be told apart, and uniqueness is the whole purpose of the key.